

Natural selection

Individual plants and animals that have a trait which helps them to survive in their particular environment are more likely to pass on that trait in their genes to the next generation. Over time, more and more individuals will come to have the trait.



Shared environment

A population of rabbits share an environment where brown and white individuals are equally likely to survive.



Blending with the scenery

They move into a snowy environment where eagles pick out and kill brown rabbits. The white ones blend with their surroundings and are therefore harder to spot.

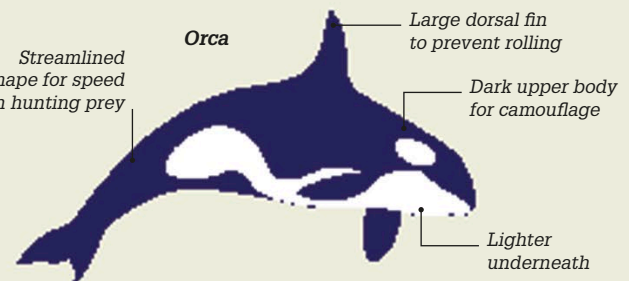


New population

More white rabbits survive than brown. After many generations, the gene that determines the white fur color comes to predominate.

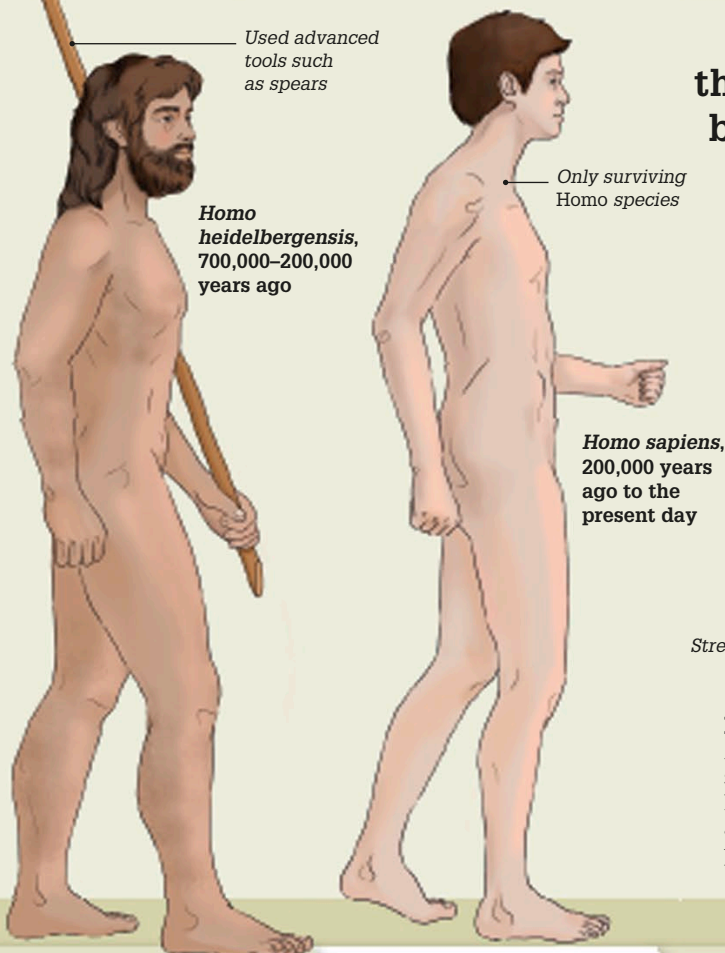
“As new species... are formed through natural selection, others will become rarer, and finally extinct.”

Charles Darwin, *On the Origin of Species*



Different but similar

Although killer whales (which are mammals) and great white sharks (which are fish) are completely different species, they have evolved similar characteristics. This is because they fill a similar role in the ocean environment. Biologists call this process “convergent evolution.”



Used advanced tools such as spears

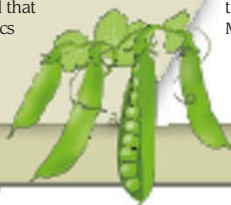
Homo heidelbergensis,
700,000–200,000 years ago

Only surviving Homo species

Homo sapiens,
200,000 years ago to the present day

1866

Austrian botanist Gregor Mendel published his study of pea plants in which he showed that certain characteristics are passed on from one generation to another.



1909

Danish botanist Wilhelm Johannsen was the first person to use the term “gene” to describe Mendel’s basic unit of heredity.

1953

American scientist James Watson and English scientist Francis Crick demonstrated that the structure of DNA, the material that carries genetic information (see pp.198–199), is a double helix (spiral).

2003

The full sequencing of the 3 billion base pairs that make up the human genome (the complete set of genes) was published.